



Title

scplotmulti — Synthetic Control Methods with Multiple Treated Units Plots.

Syntax

```
scplot , [scest uncertainty(string) uncertainty(string) joint yscalefree
xscalefree dots_tr_col(colorstyle) dots_tr_symb(symbolstyle)
dots_tr_size(markersizestyle) dots_sc_col(colorstyle) dots_sc_symb(
symbolstyle) dots_sc_size(markersizestyle) line_tr_col(colorstyle)
line_tr_patt(linepatternstyle) line_tr_width(linewidthstyle)
line_sc_col(colorstyle) line_sc_patt(linepatternstyle)
line_sc_width(linewidthstyle) spike_sc_col(colorstyle)
spike_sc_patt(linepatternstyle) spike_sc_width(linewidthstyle)
gphoptions(string) gphcombineoptions(graph combine) gphsave(string)
savedata(dta_name) precision(string) keepsingleplots pypinocheck]
```

Description

scplot implements several Synthetic Control (SC) plots even in the presence of multiple treated units and staggered adoption. The command is designed to be called after **scest** or **scpi** which implement estimation and inference procedures for SC methods using least squares, lasso, ridge, or simplex-type constraints according to [Cattaneo, Feng, and Titiunik \(2021\)](#) and for a single treated unit and staggered adoption. The command is a wrapper of the companion Python package. As such, the user needs to have a running version of Python with the package installed. A tutorial on how to install Python and link it to Stata can be found [here](#).

Companion R and Python packages are described in [Cattaneo, Feng, Palomba and Titiunik \(2025\)](#).

Companion commands are: **scdata** for data preparation, **scest** for estimation procedures, and **scpi** for inference procedures.

Related Stata, R, and Python packages useful for inference in SC designs are described in the following website:

<https://nppackages.github.io/scpi/>

For an introduction to synthetic control methods, see [Abadie \(2021\)](#) and references therein.

Options

Type of Plot

scest if specified **scplot** must be called after **scest**. Otherwise, it is presumed that **scplot** is called after **scpi**.

uncertainty(string) specifies which prediction intervals are plotted. It does not affect the plot if **scest** is specified. Options are:

insample prediction intervals quantify only in-sample uncertainty.

gaussian prediction intervals quantify in-sample and out-of-sample uncertainty using conditional subgaussian bounds.

ls prediction intervals quantify in-sample and out-of-sample uncertainty imposing a location-scale model.

qreg prediction intervals quantify in-sample and out-of-sample uncertainty using quantile regressions.

ptype(string) specifies the type of plot to be produced. If set to 'treatment', then treatment effects are plotted. If set to 'series' (default), the actual and synthetic time series are reported.

joint if specified simultaneous prediction intervals are included in the plot(s).

Scale Options

yscalefree if specified each graph has its own scale for the y axis.
xscalefree if specified each graph has its own scale for the x axis.

Marker Options

These options let the user specify color, size, and form of the markers in the plot.

dots_tr_col(*colorstyle*) specifies the color of the markers for the treated unit.
dots_tr_symb(*symbolstyle*) specifies the form of the markers for the treated unit.
dots_tr_size(*markersizestyle*) specifies the size of the markers for the treated unit.
dots_sc_col(*colorstyle*) specifies the color of the markers for the SC unit.
dots_sc_symb(*symbolstyle*) specifies the form of the markers for the SC unit.
dots_sc_size(*markersizestyle*) specifies the size of the markers for the SC unit.

Line Options

These options let the user specify color, pattern, and width of the lines in the plot.

line_tr_col(*colorstyle*) specifies the color of the line for the treated unit.
line_tr_patt(*linepatternstyle*) specifies the pattern of the line for the treated unit.
line_tr_width(*linewidthstyle*) specifies the width of the line for the treated unit.
line_sc_col(*colorstyle*) specifies the color of the line for the SC unit.
line_sc_patt(*linepatternstyle*) specifies the pattern of the line for the SC unit.
line_sc_width(*linewidthstyle*) specifies the width of the line for the SC unit.

Bar Options

These options let the user specify color, pattern, and width of the bar (spikes) in the plot. These options do not have effect if **scest** is specified.

spike_sc_col(*colorstyle*) specifies the color of the bars for the SC unit.
spike_sc_patt(*linepatternstyle*) specifies the pattern of the bars for the SC unit.
spike_sc_width(*linewidthstyle*) specifies the width of the bars for the SC unit.

Others

gphoptions(*string*) specifies additional options to modify individual plots.
gphcombineoptions(*graph combine*) specifies additional options to modify the final combined plot.
gphsave(*string*) specifies the path and the name of the .gph file that is saved by the command.
savedata(*dta_name*) saves a *dta_name.dta* file containing the processed data used to produce the plot.
precision(*string*) specifies the storage type for numeric variables generated by **scplotmulti** for plotting. Options are **double** (default) and **single**.
precision(*single*) uses Stata's default generated numeric storage type, matching earlier package behavior.
keepsingleplots) if specified saves the individual plots in .gph format in the current directory.
pypinocheck) if specified avoids to check that the version of *scpi_pkg* in Python is the one required by **scplot** in Stata. When not specified performs the check and stores a macro called to avoid checking it multiple times.

Example: Germany Data

```
Setup
. use scpi_germany.dta
```

```
Prepare data
. scdata gdp, dfname("python_scddata") id(country) outcome(gdp) time(year)
  treatment(status) cointegrated

Estimate Synthetic Control with a simplex constraint and quantify uncertainty
. scpi, dfname("python_scddata") name(simplex) u_missp

Plot Synthetic Control Estimate with Prediction Intervals
. scplot, gphsave("plot_scpi")
```

References

- Abadie, A. 2021. Using synthetic controls: Feasibility, data requirements, and methodological aspects. *Journal of Economic Literature*, 59(2), 391-425.
- Cattaneo, M. D., Feng, Y., and Titiunik, R. 2021. Prediction intervals for synthetic control methods. *Journal of the American Statistical Association*, 116(536), 1865-1880.
- Cattaneo, M. D., Feng, Y., Palomba, F., and Titiunik, R. 2025. scpi: Uncertainty Quantification for Synthetic Control Methods. *Journal of Statistical Software*, 113(1), 1-38.
- Cattaneo, M. D., Feng, Y., Palomba, F., and Titiunik, R. 2027. Uncertainty Quantification in Synthetic Controls with Staggered Treatment Adoption. *Review of Economics and Statistics*, forthcoming.

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